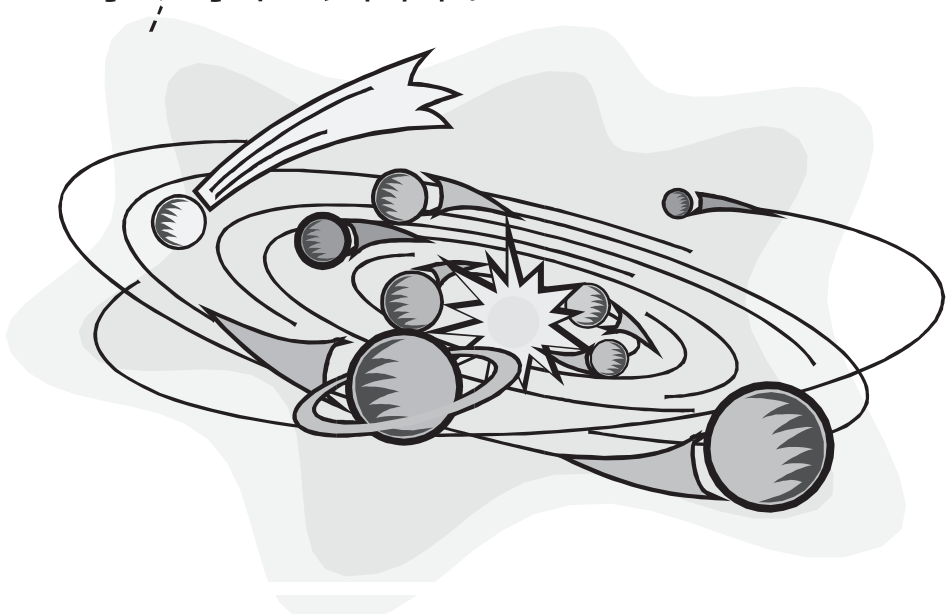
The Solar System

The solar system is made up of many parts. The Moon revolves around the Earth and the Earth revolves around the Sun. The inner solar system includes Mercury, Venus, Earth, and Mars, while the outer solar system includes Jupiter, Saturn, Uranus, and Neptune. Our solar system also has many stars and constellations. The Earth is part of a galaxy, or group of stars held together by gravity, called the Milky Way. All of these parts work together to make our solar system.



Word Power

Rotation - a complete spin on an axis.

Revolution - one complete trip around the Sun.

Lunar eclipse - a blocking of the view of the Moon when the full Moon passes through the Earth's shadow.

Solar eclipse - a blocking of the view of the Sun when Earth passes through the Moon's shadow.

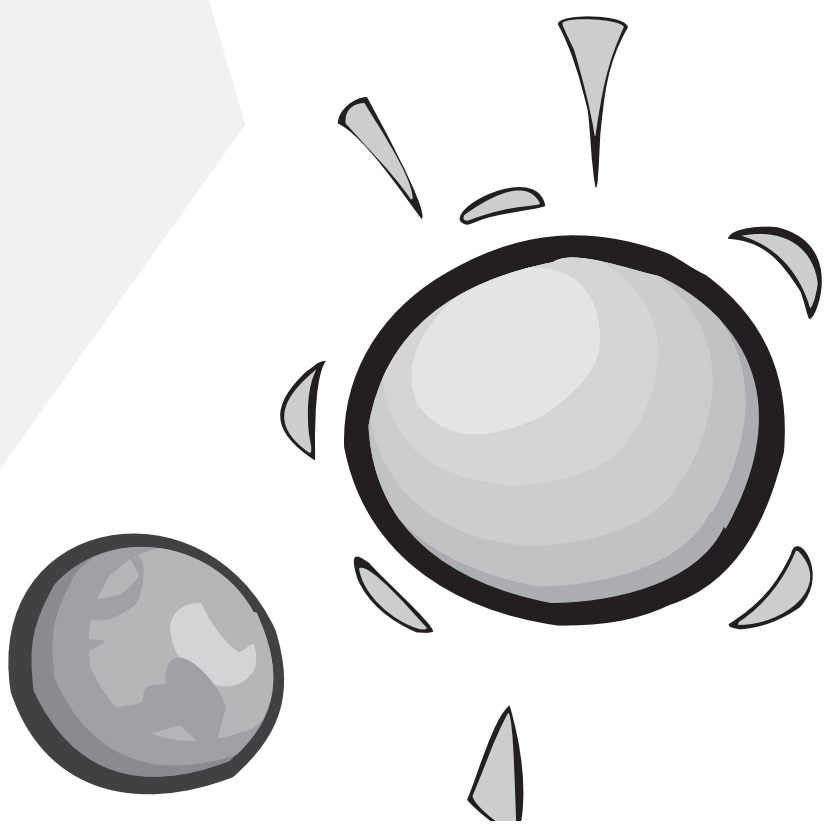
Solar system - the Sun and all of the planets, moons, and other bodies traveling around it.

Star - a large, hot ball of gas which is held together by gravity and gives off its own light.

Galaxy - large group of stars held together by gravity.

The Earth and Sun

Earth's movement makes daytime and seasons possible. Earth rotates on its axis. The axis is an imaginary line through Earth from the North Pole to the South Pole. It takes 24 hours for Earth to make one rotation. The Sun's position in the sky changes during the day. As the day progresses, the Sun appears to rise in the east and set in the west. The Sun reaches its highest point in the sky at noon.



1. Earth rotates on its O_____.

2. The axis is an imaginary line from the N_____ Pole to the S_____ Pole.

3. It takes 24 hours for Earth to make one r_____.

4. The S_____ position changes throughout the day.

5. The Sun appears to rise in the E_____ and to sets in the W_____.

Word Bank

axis	**rotation**	**west**	**Sun's**
north	**south**	**east**	

Fill in the blanks with the correct word.

1. What does the Earth rotate on? _____.

2. How long does it take for the Earth to make one rotation?

_____.

3. In what direction does the Sun rise? _____.

4. At what time is the Sun at its highest point?_____.

5. In what direction does the Sun set?_____.

Word Bank

axis	**noon**	**east**
west	**24 hours**	

1. What does the Earth rotate on? _____.
axis

2. How long does it take for the Earth to make one rotation?

_____.
24 hours

3. In what direction does the Sun rise? _____.
east

4. At what time is the Sun at its highest point?_____.
noon

5. In what direction does the Sun set?_____.
west

Word Bank

axis	**noon**	**east**
west	**24 hours**	

Example:

S ummer is my favorite time of the year
U nless it's raining. Then I can't play in my
N eighborhood with my friends.

E _____

A _____

R _____

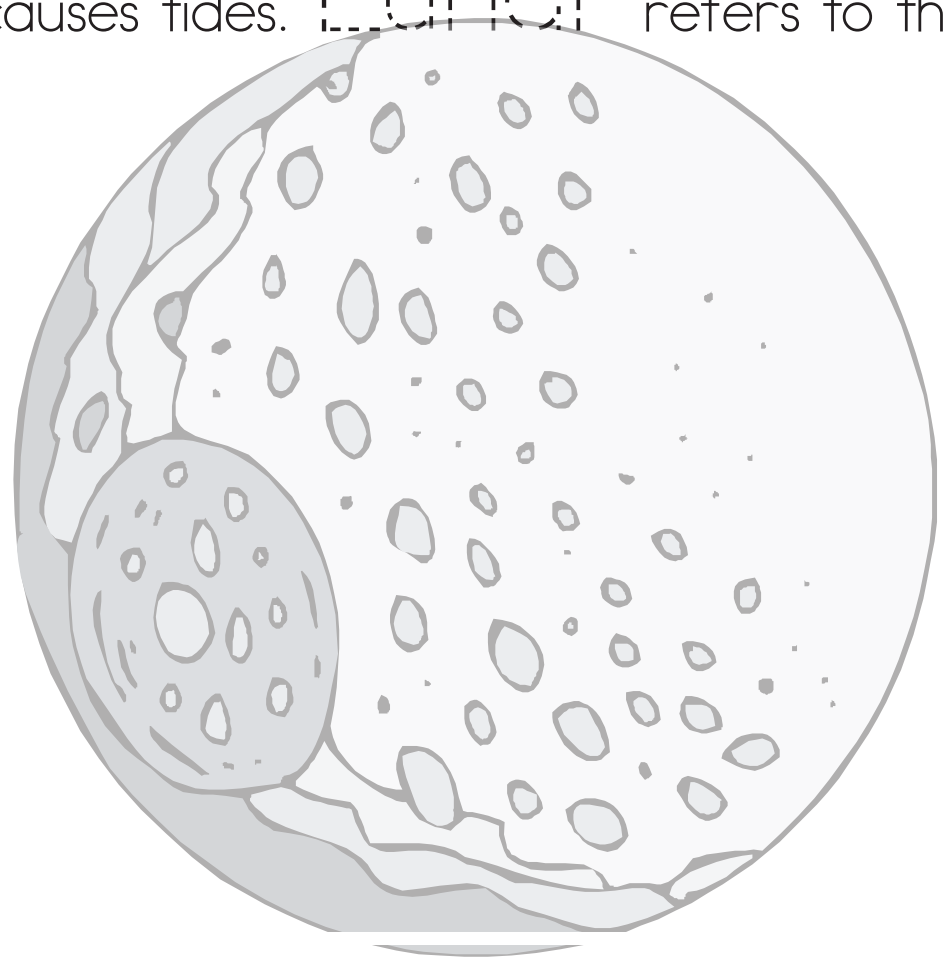
T _____

H _____

Your Poem

Write an acrostic poem about Earth.

The Moon revolves around Earth and Earth revolves around the Sun. Half of the Moon is always lit by the Sun. However, you can't always see that entire half. A lunar eclipse happens when the Moon passes into the Earth's shadow, blocking the Sun from the Moon. A solar eclipse occurs when Earth passes through the Moon's shadow, blocking the view of the Sun from Earth. Tide is the regular rise and fall of the water along a shore. The Moon causes tides. Lunar refers to the Moon.



The Moon

Read and trace the words.

1. The M_____ revolves around the Earth.

2. H_____ of the moon is always lit by the Sun.

3. A _____ eclipse occurs when
E_____ passes through the Moon's shadow
and blocks the Sun from the Moon.

4. A S_____ eclipse happens when
E_____ passes through the Moon's shadow
and blocks our view of the Sun.

5. T _____ is the regular rise and fall of the water
along a shore.

6. The M_____ causes tides.

Word Bank

Moon	**solar**	**Moon**
half	**earth**	
lunar	**tide**	

The Moon

Fill in the blanks with the correct words.

_____1. The Moon revolves around the Sun.

_____2. Half of the Moon is always lit by the Sun.

_____3. A lunar eclipse is when Earth passes through the Moon's shadow, blocking our view of the Sun.

_____4. A solar eclipse is when the Moon passes into Earth's shadow, blocking the Sun from the Moon.

_____5. A tide is the regular rise and fall of the water along a shore.

_____6. The moon causes tides.

The Moon

Answer True or False.

_____1. The Moon revolves around the Sun.
False

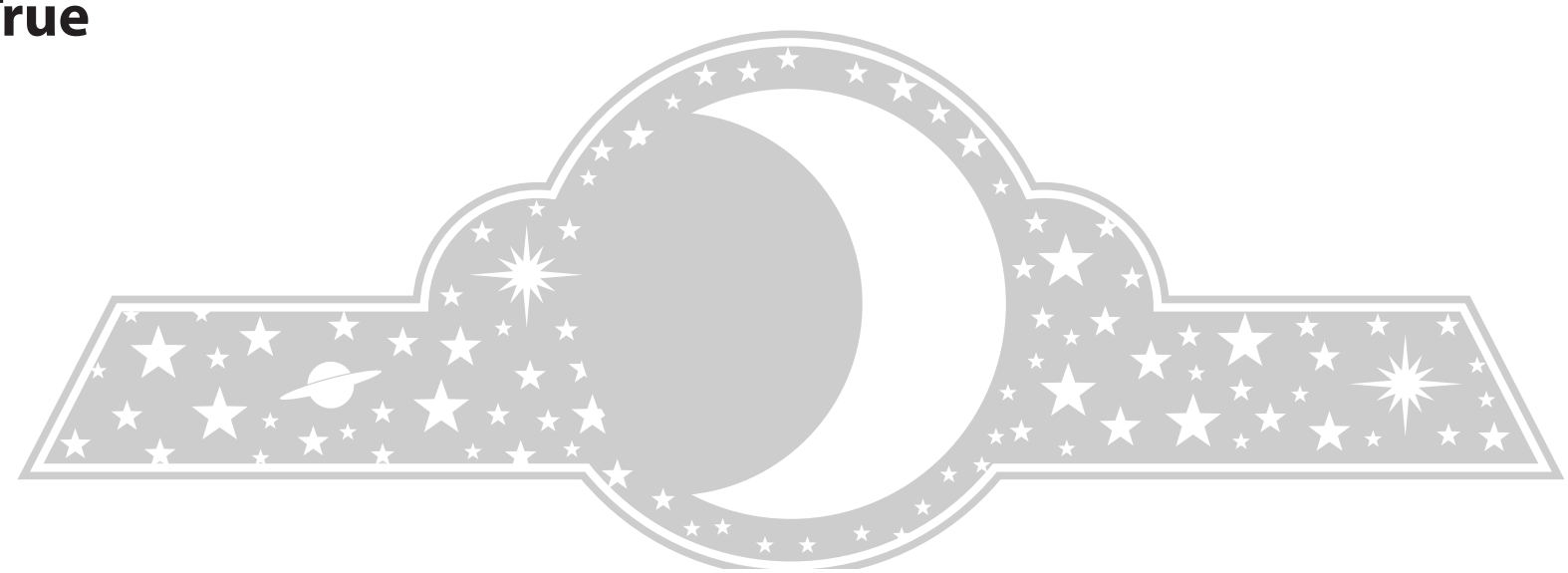
_____2. Half of the Moon is always lit by the Sun.
True

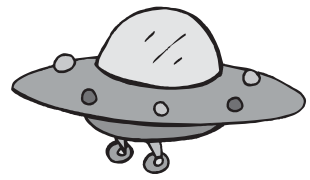
_____3. A lunar eclipse is when Earth passes through the Moon's shadow, blocking our view of the Sun.
False

_____4. A solar eclipse is when the Moon passes into Earth's shadow, blocking the Sun from the Moon.
False

_____5. A tide is the regular rise and fall of the water along a shore.
True

_____6. The moon causes tides.
True





R	P	O	L	A	V	T	I	D	E
F	E	H	J	K	M	X	W	A	W
S	L	V	H	O	V	B	R	N	G
H	W	E	O	S	D	T	T	Y	H
A	H	N	S	L	H	D	S	F	G
D	S	S	H	U	V	E	O	V	T
O	G	U	S	N	D	E	L	H	H
W	S	N	H	A	W	T	A	Y	G
Z	X	C	V	R	B	K	R	L	I
E	C	L	I	P	S	E	A	Q	L

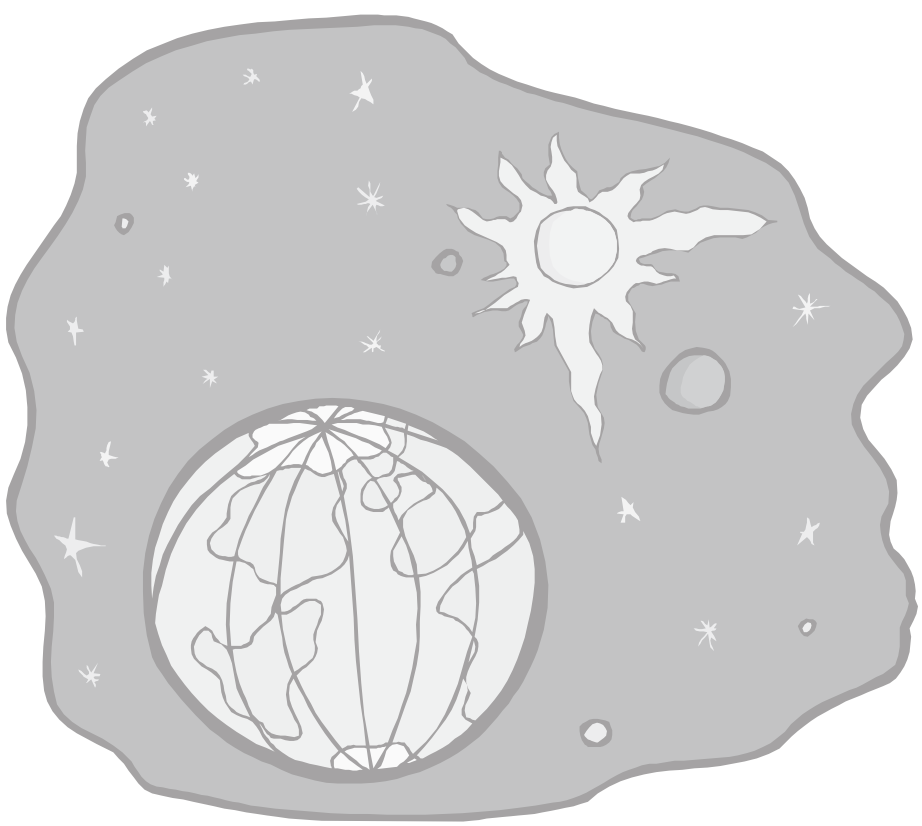
Word Bank

Moon	**Sun**	**Earth**	**lunar**	**solar**
eclipse	**tide**	**shadow**	**light**	**revolve**

The Moon

Find all the hidden words.

The solar system includes the Sun, planets, moons, and other bodies traveling in it. The inner solar system includes Mercury, Venus, Earth, and Mars. Mercury is the closest planet to the Sun. Venus rotates in a direction opposite that of the Earth. Mars has a rotation period of 24.6 hours and also has seasons. Asteroids, which are rocky and metallic objects, are found mostly between Mars and Jupiter.



The Inner Solar System

Read and trace the dotted words.

1. The inner solar system includes M_____, V_____, E_____, and M_____.

2. Mercury is C_____ to the Sun.

3. Venus r_____ in the opposite direction than of E_____.

4. Mars has a rotation period of 24.6 h_____ and also has S_____.

5. An asteroid is a rocky metallic O_____ that orbits the Sun.

6. A_____ are mostly found in between Mars and J_____.

The Inner Solar System

Complete the words.

1. The solar system is the Sun, 8 planets, _____, and other bodies traveling in it.

trees moons stars

2. The inner solar system includes Mercury, _____, Earth, and Mars.

Venus Neptune Sun

3. Mercury is the closest planet to the _____.

Moon Earth Sun

4. Venus rotates in a direction _____ of Earth's rotation.

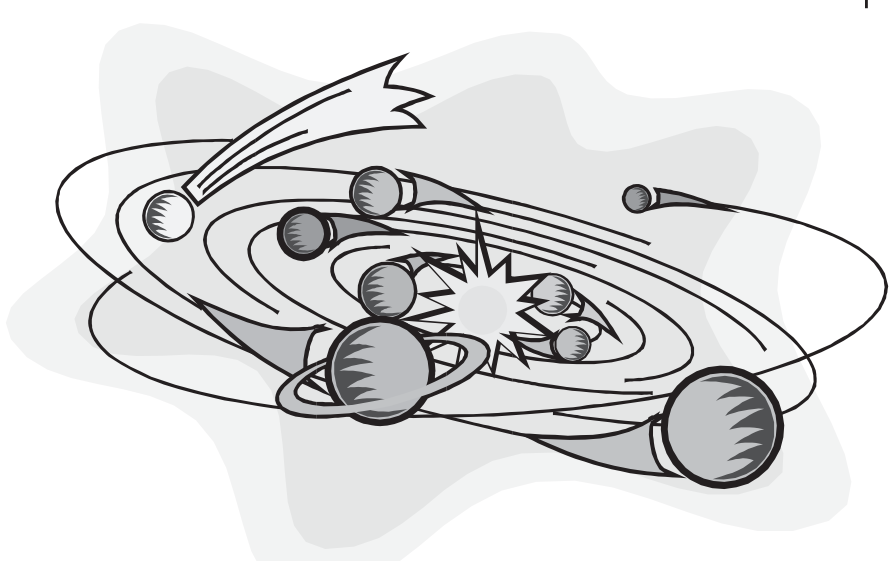
opposite same side-ways

5. Mars rotates in 24.6 hours and also has _____.

people volcanoes seasons

6. Asteroids are mostly found between _____ and Jupiter.

Mars Earth Neptune



The Inner Solar System

Circle the correct answer.

1. The solar system is the Sun, 8 planets, _____, and other bodies traveling in it.

trees moons stars

2. The inner solar system includes Mercury, _____, Earth, and Mars.

Venus Neptune Sun

3. Mercury is the closest planet to the _____.

Moon Earth Sun

4. Venus rotates in a direction _____ of Earth's rotation.

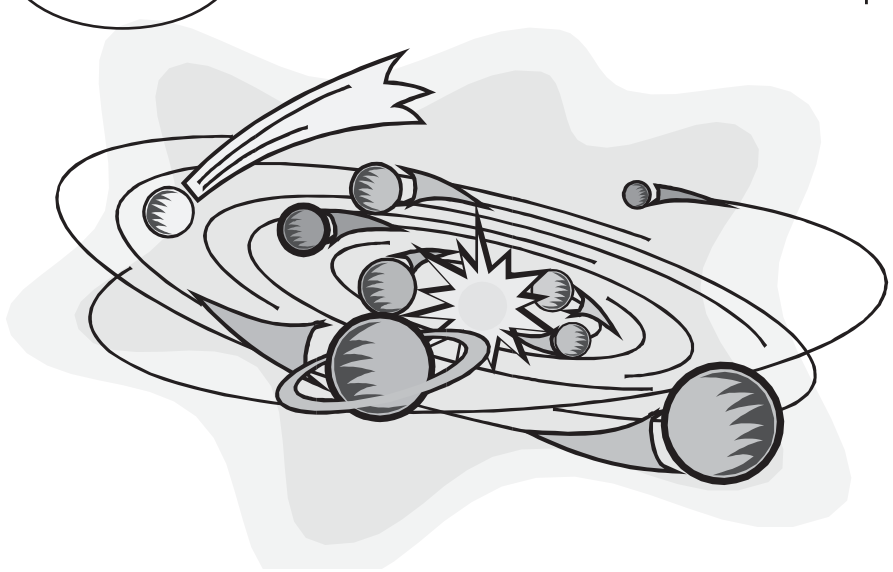
opposite same side-ways

5. Mars rotates in 24.6 hours and also has _____.

people volcanoes seasons

6. Asteroids are mostly found between _____ and Jupiter.

Mars Earth Neptune



What would your name be? _____

How many moons would you have? _____

Would you be colder or hotter than Earth? Why?

_____

_____

How long would your day be?

_____

What would your seasons be?

_____

Which holidays would you have?

_____

_____

Would you have more land or more water? Why?

_____

_____

Who or what would live on your planet?

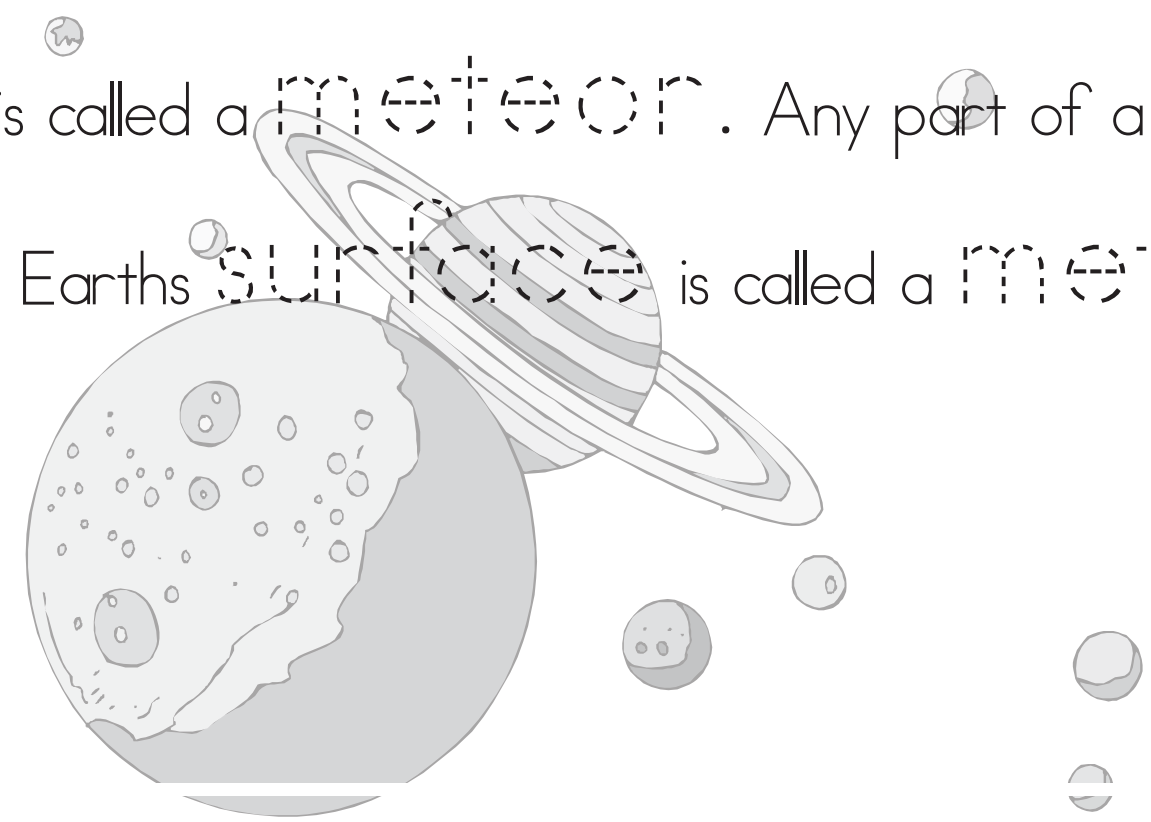
_____

_____

If I were a planet...?

Read the questions below and respond as if you were a planet.

The outer solar system includes Jupiter, Saturn, Uranus and Neptune. Pluto was once counted as the ninth planet, but is now considered to be a "dwarf planet." The four outer planets are larger than the inner planets and tend to rotate faster. A comet is a ball of ice and rock that orbits the Sun. There are also small asteroids, which are called meteoroids. When a meteoroid hits Earth's atmosphere, it begins to burn. At this point it is called a meteor. Any part of a meteoroid that reaches Earths surface is called a meteorite.



The Outer Solar System

Read and trace the words.

1. The O_____ solar system includes Jupiter, Saturn, Uranus and N_____. Pluto is now called a d_____ P_____ instead of the ninth planet.

2. The four outer planets are larger than the I_____ planets and rotate I_____.

3. A C_____ is a ball of I_____ and rock that orbits the Sun.

4. Small asteroids are called M_____.

5. When a meteoroid hits Earth's atmosphere, it is called a m_____.

6. Any part of a meteoroid that reaches the Earth's s_____ is called a m_____.

Word Bank

outer	**comet**	**meteoroids**	**meteor**	**meteorite**
inner	**faster**	**Neptune**	**surface**	**ice**
dwarf planet				

The Outer Solar System

Read and trace the words.

1. What are the planets that are in the outer solar system?

_____

_____

2. How are the outer planets different from the inner planets?

_____

_____

3. What do comets orbit? _____

4. What are small asteroids called? _____

5. What is a meteoroid called when it hits the Earth's atmosphere?

_____

6. If any part of a meteoroid reaches Earth's surface, what is it called? _____

7. Write the name of the dwarf planet. _____

1. What are the planets that are in the outer solar system?

_____

_____

Jupiter, Saturn, Uranus and Neptune

2. How are the outer planets different from the inner planets?

_____

_____

They are larger than the inner planets and rotate faster.

3. What do comets orbit? _____

the Sun

4. What are small asteroids called? _____

Meteoroids

5. What is a meteoroid called when it hits the Earth's atmosphere?

_____

Meteor

6. If any part of a meteoroid reaches Earth's surface, what is it called? _____

Meteorite

7. Write the name of the dwarf planet. _____

Pluto

1. struna_____

2. narsuu_____

3. itjupre_____

4. outlp_____

5. etmoc_____

6. rdiosoeetm_____

7. eeortm _____

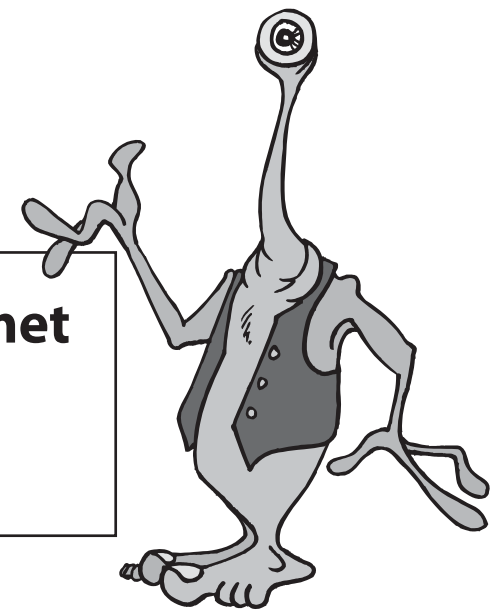
8. tieteoerm_____

9. uns_____

10. etlanps_____

Word Bank

Saturn	**Uranus**	**Jupiter**	**comet**
meteoroids	**meteorite**	**Pluto**	
meteor	**planets**	**Sun**	



The Outer Solar System

Unscramble the words.

A star is a large, hot ball of gas held together by gravity, and gives off its own light. A constellation is a group of stars that appears to form a pattern. Magnitude is used to describe the brightness of a star. A star begins its life as a nebula, an enormous cloud of gas and dust in space. During its life, a star may become unstable; it may develop into a supernova—a star that explodes. A large supernova is called a black hole, or an object whose gravity is so strong that light cannot escape it. The Sun is the closest star to Earth. There are billions of stars throughout the universe.

Stars

Read and trace the words.

1. A S_____ is a large, hot ball of gas held together by G_____.

2. A constellation is a group of stars that appear to form a P_____.

3. Magnitude is used to describe the b_____ of a star.

4. A star begins its life as a n_____.

5. During its life a star may become U_____; it may develop into a S_____ – a star that explodes.

6. A b_____ hole is an object whose gravity is so strong that light cannot escape it.

7. The S_____ is the closest star to Earth.

Word Bank

Sun	**black**	**pattern**	**gravity**	**supernova**
brightness	**nebula**	**unstable**	**star**	

Stars *Fill in the blanks.*

_____1. A star is a small asteroid.

_____2. A constellation is a group of stars that forms a pattern.

_____3. Magnitude is used to describe the brightness of a star.

_____4. A star begins its life as a supernova.

_____5. An unstable star is called the Sun.

_____6. A black hole is an object whose gravity is so strong that light cannot escape it.

_____7. The Sun is the closest star to Earth.

_____8. There are just 1,000 stars in the universe.

Stars

Answer True or False.
If False, write the correct answer.

False 1. A star is a small asteroid.
large hot ball of gas

True 2. A constellation is a group of stars that forms a pattern.

True 3. Magnitude is used to describe the brightness of a star.

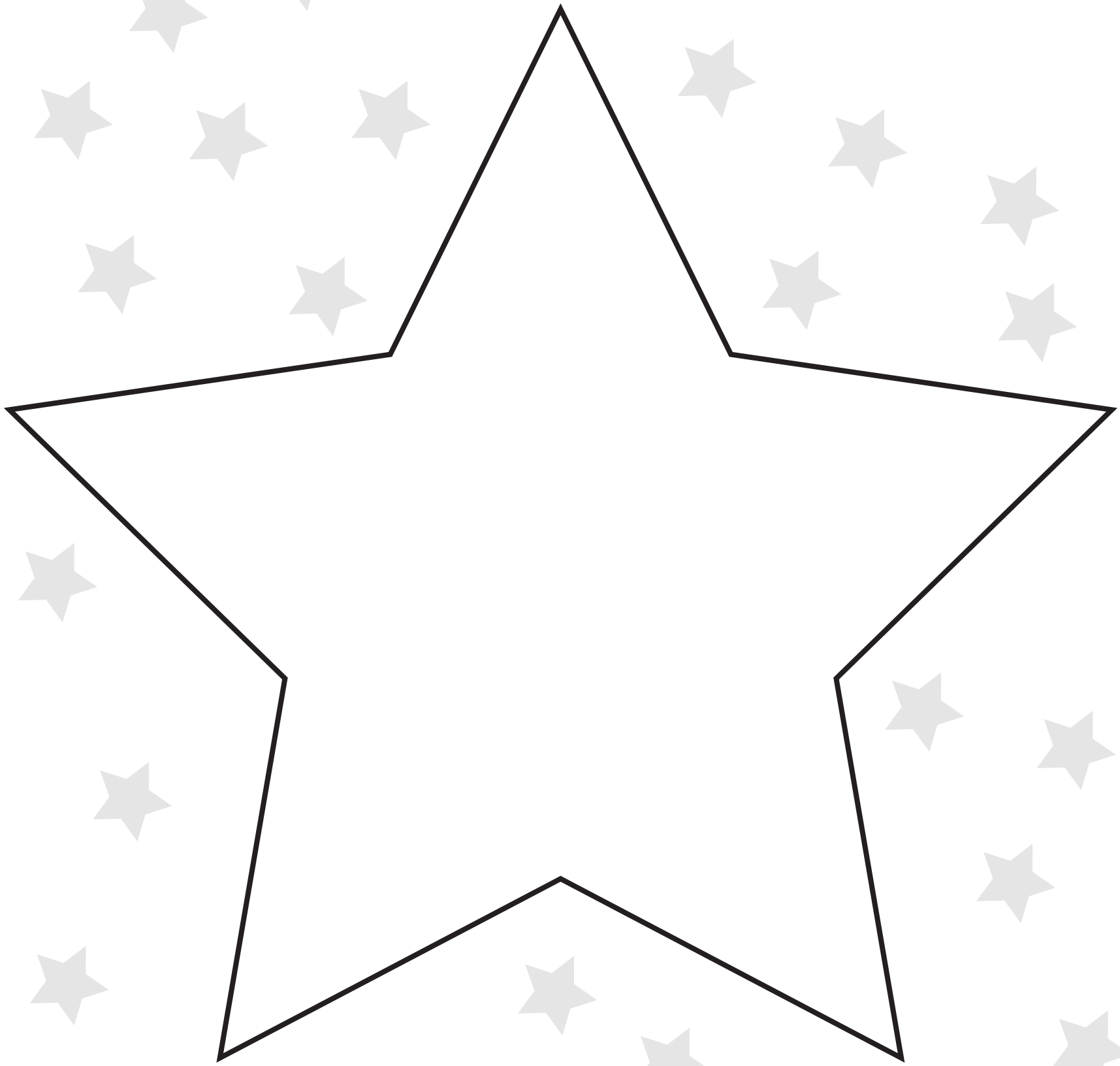
False 4. A star begins its life as a supernova.
nebula

False 5. An unstable star is called the Sun.
supernova

True 6. A black hole is an object whose gravity is so strong that light cannot escape it.

True 7. The Sun is the closest star to Earth.

False 8. There are just 1,000 stars in the universe.
billions



Stars

Decorate the star to reflect your personality.

Groups of stars are called galaxies. A galaxy is a large group of stars held together by gravity. The stars orbit the center of their galaxy. A spectrum is a band of colors made when white light is broken up. As space expands, the wavelength of light traveling through space also expands. This increase in wavelength is called an expansion redshift. Astronomers have discovered a form of electromagnetic radiation coming from all directions in space. This radiation is left over from the beginning of the universe and is called background radiation. A quasar is an extremely bright, extremely distant high-energy source.

1. A G_____ is a large group of stars held together by G_____.

2. Stars orbit the C_____ of their galaxies.

3. A S_____ is a band of colors made when white light is broken up.

4. The increase in wavelength of light due to space expanding is called an E_____ redshift.

5. B_____ radiation is left over from the beginning of the universe.

6. A Q_____ is an extremely bright, extremely distant, high-energy source.

Word Bank

expansion	**background**	**center**	**quasar**
galaxy	**gravity**	**spectrum**	

1. A group of stars held together by gravity is called a

_____.

2. Stars orbit the _____ of their galaxy.

3. A band of colors made when white light is broken up is called a

_____.

4. The increase in wavelength of light due to space expanding is called

an _____ redshift.

5. _____ radiation is left over from the

beginning of the universe.

6. A _____ is an extremely bright, extremely

distant, high-energy source.

1. A group of stars held together by gravity is called a

_____.
galaxy

2. Stars orbit the _____ of their galaxy.
center

3. A band of colors made when white light is broken up is called a

_____.
spectrum

4. The increase in wavelength of light due to space expanding is called

an _____ redshift.
expansion

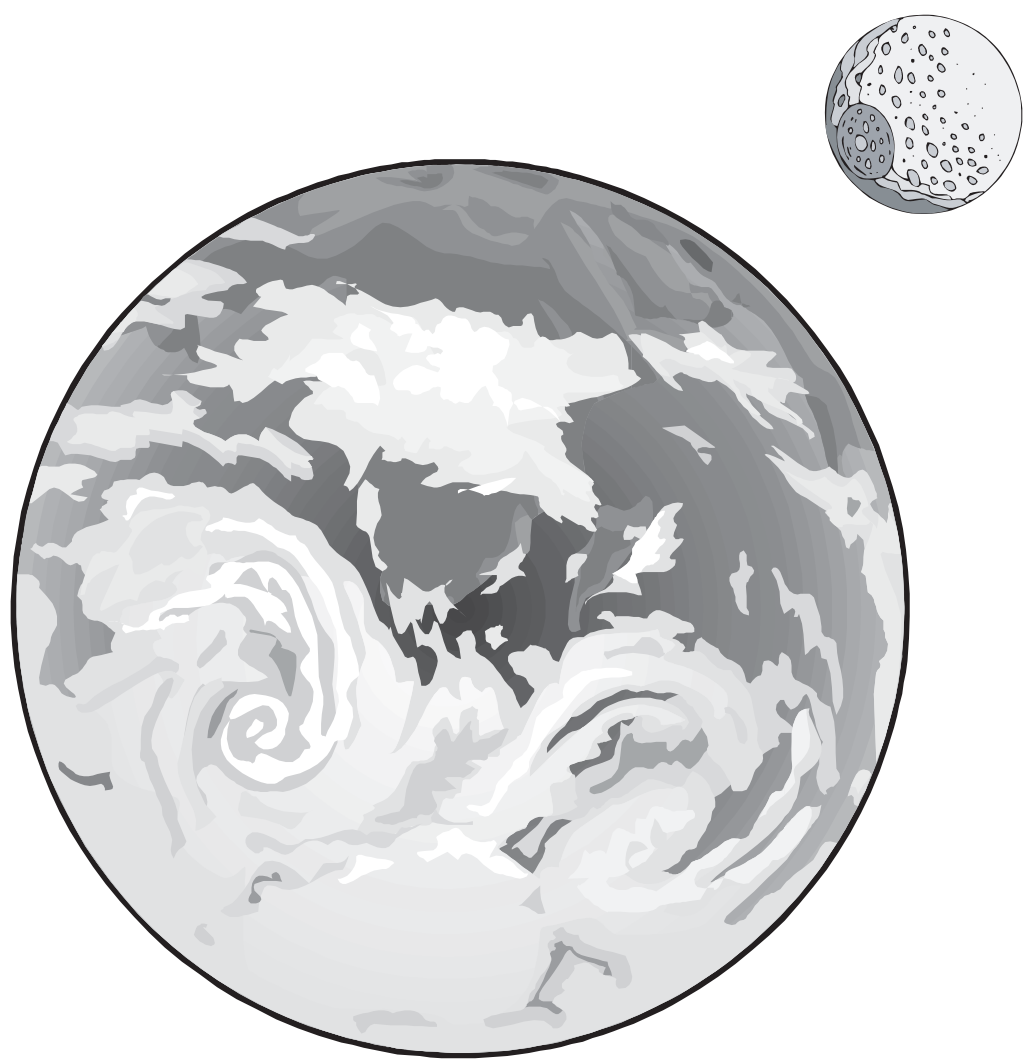
5. _____ radiation is left over from the
background

beginning of the universe.

6. A _____ is an extremely bright, extremely
quasar

distant, high-energy source.

The name of my galaxy is _____



Galaxies and Beyond

Draw your own galaxy using stars and draw the spaceship that would take you there.

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